SANJAY PADMARAJ

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EDUCATION

New Horizon College of Engineering, Bangalore **2025 2029 (expected)**
Bachelor of Engineering in Computer Science & Engineering CGPA: **9.75 / 10**

TECHNICAL SKILLS

Proficient: Python, C, SQL (MySQL)
Familiar: Java, HTML/CSS
AI / ML: RAG, LLM and multi-agent apps (LangGraph), Hugging Face, Sentence-Transformers, scikit-learn, Pandas, NumPy
Tools: GitHub, Docker

PROJECTS

Kotlex **Multi-agent Marketing Campaign Builder (HACKERING 2.0)** GitHub
Python, FastAPI, LangGraph, Server-Sent Events, Tesseract OCR, pytest

- Built a round table of **12 LLM agents** that turns a caf owner's voice note (Kannada, Hindi or English) into multilingual copy, a poster and a reel, orchestrated with LangGraph and streamed live to the browser.
- Wrote a code-only Fact Guard that blocks drafts contradicting owner-confirmed prices and timings, a dependency graph that regenerates only affected assets, and an LLM router with fallback; **500+ tests** in CI.

CampusOS **Unified Campus-Life Platform** GitHub
React 18, TypeScript, Vite, Tailwind CSS, TanStack Query, Supabase (PostgreSQL, Auth, Edge Functions)

- Built a full-stack platform consolidating **six campus workflows** (events, clubs, community feed, hackathon team-matching, resource booking, issue reporting) previously spread across chat groups, forms and portals.
- Designed the PostgreSQL schema and Supabase backend with Razorpay order and webhook handling in Edge Functions; set up a **five-stage GitHub Actions pipeline** (lint, typecheck, unit, build, E2E) with Jest and Playwright.

AI WordPress Builder **Chat-driven Website Builder** GitHub
TypeScript, Docker Compose, WP-CLI, Playwright, LLMs

- Built an agent that turns a chat brief into a structured site spec, then builds a real WordPress site (theme, pages, navigation, plugins) via WP-CLI inside an isolated per-project Docker Compose stack.
- Routed every tool call through one dispatcher enforcing permission tiers, allowlists and audit logging; added checkpointed edits with undo (database restore) and automated Playwright smoke tests.

C Visualizer **Interactive C Learning Tool** GitHub
Next.js, TypeScript, Tailwind CSS, CodeMirror, Vitest, Playwright

- Wrote a **C interpreter in TypeScript** that runs in the browser and visualises arrays, pointers, structs, stack and heap step by step, with a trace table and plain-English explanations.
- Added live syntax-error detection with one-click fixes and AI autocomplete with fallback across four LLM providers; verified interpreter output against gcc in Vitest.

Aero Engine Digital Twin **Smart India Hackathon Prototype** GitHub
Node.js, Express, Socket.IO, three.js

- Simulated telemetry for a 3-UAV fleet (13 sensors per engine, 9 injectable faults) streamed every 2 s to a live dashboard and 3D engine model; predicted faults from rolling z-scores to output a 0 100 health score and remaining-useful-life estimate.

Also built: Smart Bandage (FastAPI + React Native biosensor stack, 159 tests) Club Hub (Next.js club manager with QR tickets & waitlists) Accident Detection (ESP32 firmware with IMU + GPS)

EXPERIENCE & LEADERSHIP

STEM Club, New Horizon College of Engineering Website & AI Chatbot Developer **2025 2026**

- Built and maintain the club's official WordPress site on the college domain (newhorizonindia.edu/co-curricular/stem).
- Built and deployed a **RAG chatbot** on Hugging Face LLMs on the live site to answer student queries.

MAD Club, New Horizon College of Engineering Website Developer

- Built the club's official website.

International IEEE Conference (ICEMCSI) Host **June 2026**

PATENT, ACHIEVEMENTS & CERTIFICATIONS

- **Provisional patent filed:** Smart Blind Stick AI + IoT assistive cane with obstacle detection and user & caregiver apps.
- **1st place:** E=mc^2 STEM Expo 2026; UDAAN Business Plan Competition 2026; Algo Relay (NHCE MCA Dept.)
- **2nd place:** QuantumX Intercollege Hardware Tech Expo 2026 **3rd place:** Nexgen Convergence Hardware Hustle Tech Expo
- **Qualified:** Smart India Hackathon (college-level round) MSME Hackathon
- **Certifications:** Data Analytics (IIT Madras) AI in Physics (IIT Madras) C Programming (IIT Bombay)